

Stochastic Process It is a family of random variables $\{X(t) : t \in T\}$ defined on a common sample space S and indexed by the parameter t , which varies on an index set T . The values assumed by the random variables $X(t)$ are called states and the set of all possible values from the state space of the process is denoted by I . If the state space is discrete, then the stochastic process is known as a chain. In this case the state space is assumed to be $I = \{0, 1, 2, \dots\}$. Thus a finite stochastic process consists of a sequence of experiments in which each experiment has a finite number of outcomes with given probabilities.

A Markov process is a stochastic process whose entire past history is summarized in its current state. That is the ‘**future**’ is independent of its ‘**past**’.

Markov Chain: It is a Markov process in which the state space I is discrete (finite or countably infinite). Thus a finite Markov chain is a finite stochastic process consisting of a sequence of trials whose outcomes say $x_1, x_2, \dots$ satisfy the following two conditions:

- Each outcome belongs to the state space $I = \{a_1, a_2, \dots, a_m\}$, which is the finite set of outcomes.
- The outcome of any trial depends at most upon the outcome of the immediately preceding trial and not upon any other previous outcomes. This Markov property can be stated as

$$\begin{aligned} P(X_n = i_n | X_0 = i_0, X_1 = i_1, \dots, X_{n-1} = i_{n-1}) \\ = P(X_n = i_n | X_{n-1} = i_{n-1}). \end{aligned}$$

Now the system is said to be in state a_i at time n or at the n^{th} step if a_i is the outcome on the n^{th} trial. Associated with each ordered pair of states (a_i, a_j) , the number p_{ij} gives the probability that system changes from i^{th} state to j^{th} state. In other words p_{ij} is the probability that a_j occurs immediately after a_i occurs. The number p_{ij} are known as transition probabilities.

Transition matrix Here the following matrix P is the square matrix with transition probabilities p_{ij} .

$$\begin{pmatrix} p_{11} & p_{12} & \dots & p_{1n} \\ p_{21} & p_{22} & \dots & p_{2n} \\ \dots & \dots & \dots & \dots \\ p_{m1} & p_{m2} & \dots & p_{mn} \end{pmatrix}$$

Probability Vector: It is a vector with $v = (v_1, v_2, \dots, v_n)$ where the quantities $v_1, v_2, \dots, v_n$ are called the components of the vector with $\sum_{i=1}^n v_i = 1$ and $v_i \geq 0$.

Stochastic Matrix: Let P is square matrix with each row being a probability vector. In other words, all the entries of P are non negative and the sum of entries of any row is one.

A vector v is said to be a **fixed vector or a fixed point of a matrix** A if $vA = A$ and $v \neq 0$.

A stochastic matrix P is said to be regular if all the entries of some power P^m are positive.

Remark: 1. If $v = (v_1, v_2, v_3)$ is a probability vector of a stochastic matrix $\begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix}$, then vP is also a probability vector.

Proof. Consider

$$\begin{aligned} vP &= (v_1, v_2, v_3) \times \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix} \\ &= (v_1a_1 + v_2a_2 + v_3a_3, v_1b_1 + v_2b_2 + v_3b_3, v_1c_1 + v_2c_2 + v_3c_3) \end{aligned}$$

Since a_i, b_i, c_i and v_i are all nonnegative for any i , the components of vP are all nonnegative. Hence the sum of the components of vP is

$$vP = v_1(a_1 + b_1 + c_1) + v_2(a_2 + b_2 + c_2) + v_3(a_3 + b_3 + c_3) = v_1 \times 1 + v_2 \times 1 + v_3 \times 1 = 1.$$

(Since P is the stochastic matrix and v is the given probability vector.) □

2. If P and Q are stochastic matrices, then their product PQ is also a stochastic matrix. In general, P^n is a stochastic matrix for all positive integer n .

3. Let $t = \{t_1, t_2, \dots, t_m\}$ be a vector and T be a square matrix whose rows are each the same vector t . Then $pT = t$, if $p = (p_1, p_2, \dots, p_m)$ is a probability vector.

4. The transition matrix P of a Markov chain is a stochastic matrix.

5. Let P be a regular stochastic matrix, then

- P has a unique fixed probability vector t and the components of t are all positive.
- The sequence $P, P^2, \dots$ of powers of P approaches the matrix T whose rows are each the fixed point t .
- If p is any probability vector, then the sequence of vectors $pP, pP^2, pP^3, pP^4, \dots$ approaches the fixed point t .

Higher Transition Probabilities: The entry p_{ij} in the transition probability matrix P is the probability that the system moves from state a_i to state a_j in one step. That is $a_i \rightarrow a_j$.

The probability that a Markov chain will move from state i to state j in exactly n steps is denoted by p_{ij}^n . That is

$$a_i \rightarrow a_{k_1} \rightarrow a_{k_2} \rightarrow \dots \rightarrow a_{k_{n-1}} \rightarrow a_j$$

The matrix formed by the probabilities p_{ij}^n is called the n -step transition matrix denoted by P^n and obviously P^n is a stochastic matrix. Also it is observed that n step transition matrix is equal to n^{th} power of P .

Let P be a transition probability matrix and let $p = p_i = (p_1, p_2, \dots, p_m)$ be the probability distribution at some arbitrary time. Then $pP, pP^2, pP^3, \dots, pP^n$ are the probabilities of the system after one step, two step, $\dots$, n -step. Let $p^0 = (p_1^0, p_2^0, \dots, p_m^0)$ be the initial probability distribution at the start of the process and let $p^n = (p_1^n, p_2^n, \dots, p_m^n)$ denote the n^{th} step probability distribution at the end of n^{th} steps.

$$\therefore p^1 = p^0 p, \quad p^2 = p^1 p = p^0 p^2, \quad \dots, \quad p^n = p^{n-1} p = p^0 p^n.$$

Stationary distribution of regular Markov chains A Markov chain is said to be regular if the associated transition probability matrix P is regular. If P is a regular stochastic matrix, then the sequence of n -step transition matrices $P^2, P^3, \dots, P^n$ approaches the matrix T whose rows are each the unique fixed probability vector v of P .

We have

$$p^{(n)} = p^{(0)} p^n, \quad \text{where} \quad p^{(n)} = [p_1^{(n)}, p_2^{(n)}, \dots, p_m^{(n)}].$$

Further as $n \rightarrow \infty$, $p_i^{(n)} \rightarrow v_i$, where $1 \leq i \leq m$ and $(v_1, v_2, \dots, v_m) = v$ is called the stationary probability vector of the Markov chain.

Note: A Markov chain is said to be irreducible if every state can be reached from every other state in a finite number of steps *i.e.* $p_{ij}^n \geq 0$ for some $n > 1$

$\implies$ Markov chain is irreducible if the associated transition probability matrix is regular.

Absorbing State: A state a_i of a Markov chain is said to be absorbing state if the system remains in the state a_i once it enters there, *i.e.* a state a_i is absorbing if $p_{ij} = 1$ for $i = 1$ and $p_{ij} = 0$ if $i \neq j$.